

SHREY SINGALA

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A Driven Computer Engineering student at Penn State with a passion for building real systems, from FPGAs and embedded robotics to cloud infrastructure. Seeking opportunities to apply hands-on hardware and software expertise in a fast-paced engineering environment, where I can contribute strong technical depth, a collaborative mindset, and a proven ability to deliver from concept to completion.

EDUCATION

Pennsylvania State University - University Park
Bachelor's, Computer Engineering
Minor in Quantum Computing

August 2023 - May 2027
GPA: 3.62

EXPERIENCE

AWS Cloud Club

Co-Founder & Events Director

January 2025 - Present
State College, PA, USA

- Identified a gap in cloud computing resources at Penn State and co-founded a club that became one of the fastest-growing organizations, attracting over 150 active members
- Oversaw a 6-member operations team to promote events and ensured seamless execution of AWS workshops and club activities
- Coordinated with AWS Solutions Architects to design and deliver hands-on cloud computing workshops and AWS Jams, increasing member exposure to industry-standard tools and practices

CSE Learning Assistant

Data Structures & Algorithms

August 2025 - Present
State College, PA, USA

- Instructed recitation sessions for 100+ students on data structures and algorithms with tailored guidance and interactive problem-solving techniques
- Provide detailed feedback on algorithmic correctness, runtime complexity, and edge cases to help students improve problem-solving skills
- Received faculty commendation for the quality and consistency of feedback delivered throughout the semester

Student Space Program Laboratory

Student Training Program

January 2024 - May 2024
State College, PA, USA

- Designed and built a rocket payload from concept to launch as part of a semester-long design-build-fly program under Penn State's SSPL
- Instrumented payload with sensors to collect temperature, pressure, altitude, and IMU data throughout ascent
- Completed successful launch and data collection; diagnosed parachute deployment failure during post-mission debrief and identified recovery system design improvements

PROJECTS

Two-Wheeled Self-Balancing Robot (PID vs LQR)

May 2026 - Present

- Built a Segway-style balancing robot on the STM32 Nucleo F446RE with MPU-6050 IMU over I2C and N20 motors with encoders for tilt and velocity feedback
- Implemented a complementary filter for sensor fusion and bare-metal PID on hardware timer interrupts, driving motors with PWM

Multi-Threaded Chat Application

January 2026 - May 2026

- Built a full-duplex C client/server chat app over TCP with a custom binary protocol and proper network byte order handling
- Spawned a dedicated pthreads receive thread on the client, decoupling concurrent inbound I/O from user input
- Enforced server- and client-side rate limiting without self-disconnects via careful shared-state synchronization across threads

Analog Audio Signal Processing System

November 2025 - December 2025

- Designed a multi-stage analog audio processing system with stereo mixing, Baxandall tone control, and LED-based amplitude visualization
- Selected and analyzed components for op-amp summing and difference amplification to achieve frequency-selective gain shaping
- Simulated functionality in NI Multisim, then progressed through breadboard and PCB prototyping to validate real-world performance
- Diagnosed and resolved resistor ladder tuning and signal attenuation issues, improving overall system stability

2-Way Superscalar 5 Staged Pipelined CPU

October 2025 - November 2025

- Built a 2-way superscalar MIPS processor with a 5-stage pipeline on FPGA, supporting dual-issue execution with hazard detection and stall logic in Verilog
- Extended the datapath to support additional R-type instructions, enabling robust execution across a wider instruction set
- Validated cycle-accurate behavior on the Xilinx Zynq-7000 using AMD Vivado waveform simulation

SKILLS

Software Oriented: C/C++, Python, Java, MySQL, AWS, TCP/IP, Git, Data Analysis

Hardware Oriented: Verilog, FPGA, STM32, Bare-Metal Firmware, I2C, PWM, Sensor Fusion, PID/LQR Control, 3D Modeling, Analog Circuit Design, PCB Prototyping